## *Average life/mean life*

The *average life or mean life* is defined as the ratio of the total life time of all the radioactive atoms to the total numbers of such atoms in it. It is denoted by $\tau$.

$$\tau = \frac{\text{Total life of all atoms}}{\text{Total number of atoms}}$$

Suppose,

The number of atoms at the beginning is $N_0$,
At time $t$, the number of atoms is $N$,
and d$N$ atoms disintegrate between $t$ and $t+dt$.

It means that $(-\mathrm{dN})$ atoms have lived for a time $t$ seconds.

$$\text{Total life of } (-\mathrm{dN}) \text{ atoms} = -\mathrm{dN.t}$$

Since all the atoms disintegrate in time from zero to infinity, the sum of the life periods of all the atoms

$$= \int_0^\infty -\mathrm{dN.t}$$

Thus the average life

$$\tau = \frac{1}{N_0}\int_0^\infty -\mathrm{dN.t} \qquad (1)$$

The radioactive decay law is

$$N = N_0 e^{-\lambda t}$$

We can write

$$\frac{dN}{dt} = -\lambda N_0 e^{-\lambda t}$$

$$\therefore \qquad dN = -\lambda N_0 e^{-\lambda t} dt$$

Substituting the value of d$N$ in equation (1) we get

$$\tau = \frac{1}{N_0}\int_0^\infty \left(\lambda N_0 e^{-\lambda t} dt\right) t$$

$$\tau = \lambda \int_0^\infty t e^{-\lambda t} dt$$

Integrating by parts we get

$$\tau = \lambda\left[t\int e^{-\lambda t}dt - \int\left\{\frac{d}{dt}t\int e^{-\lambda t}dt\right\}dt\right]_0^\alpha$$

$$= \lambda\left[t\frac{e^{-\lambda t}}{-\lambda} - \int\frac{e^{-\lambda t}}{-\lambda}dt\right]_0^\alpha$$

$$= \lambda\left[t\frac{e^{-\lambda t}}{-\lambda} - \frac{e^{-\lambda t}}{\lambda^2}\right]_0^\infty$$

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003

$$= \lambda \left[ \frac{\left(-\lambda t e^{-\lambda t}\right) - \left(e^{-\lambda t}\right)}{\lambda^2} \right]_0^\infty$$

$$= -\frac{1}{\lambda}(0-1)$$

$$\tau = \frac{1}{\lambda}$$

This is the expression for average life.

***Relation between half-life and average life/mean life***

We know, half-life

$$T_{1/2} = \frac{0.693}{\lambda} \qquad (1)$$

Mean life

$$\tau = \frac{1}{\lambda} \qquad (2)$$

From the above two equations we can write

$$T_{1/2} = 0.693\tau$$

$$Half - life = 0.693 \times Mean\ life$$

$$Half - life = 69.3\%\ of\ Mean\ life$$

These are the Relation between half-life and mean life.

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003

## ***Mathematical Problems***

***Problem-1:*** *The disintegration constant $\lambda$ of a radioactive element is 0.00231 per day. Calculate its half-life and average life.*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$T_{1/2} = \frac{0.693}{0.00231}$$

$$\therefore \quad T_{1/2} = 300\ days\ (\boldsymbol{Ans})$$

Here,

$\lambda = 0.00231\ per\ day$

$T_{1/2} = ?$

$\tau = ?$

Mean-life

$$\tau = \frac{1}{\lambda} = \frac{1}{0.0231} = 432.9\ days\ (\boldsymbol{Ans})$$

***Problem-2:*** *How long does it take for 60.0 percent of a sample of radon to decay?*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{3.82}$$

$$\therefore \quad \lambda = 0.1814\ per\ day$$

Here,

$N = \left(1 - \frac{60}{100}\right) N_0$

$N = (1 - 0.6)N_0$

$N = 0.4N_0$

$\frac{N_0}{N} = \frac{1}{0.4} = 2.5$

$T_{1/2} = 3.82\ days$

$t = ?$

We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln\frac{N_0}{N} = \lambda t$$

$$t = \frac{1}{\lambda}\ ln\frac{N_0}{N}$$

$$t = \frac{1}{0.1814} ln\, 2.5$$

$$\therefore \quad t = 5.05\ days\ (\boldsymbol{Ans})$$

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003

***Problem-3:*** *How long will it take for a sample radium D to decrease to 10%, if its half-life is 22 years?*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{22}$$

$$\therefore \quad \lambda = 0.0315 \; per \; year$$

Here,

$$N = \frac{10}{100} N_0$$

$$\frac{N_0}{N} = 10$$

$$T_{1/2} = 22 \; years$$

$$t = ?$$

We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln \frac{N_0}{N} = \lambda t$$

$$t = \frac{1}{\lambda} \; ln \frac{N_0}{N}$$

$$t = \frac{1}{0.0315} ln \, 10$$

$$t = 73.11 \; years \; (\boldsymbol{Ans})$$

***Problem-4:*** *Calculate the time in which the activity of a sample of thorium reduces to 90% of its original value. Assume the half-life of thorium to be* $1.4 \times 10^{10}$ *years.*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{1.4 \times 10^{10}}$$

$$\therefore \quad \lambda = 4.95 \times 10^{-11} \; per \; year$$

Here,

$$N = \frac{90}{100} N_0$$

$$\frac{N_0}{N} = \frac{10}{9}$$

$$T_{1/2} = 1.4 \times 10^{10} \; years$$

$$t = ?$$

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003

We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln\frac{N_0}{N} = \lambda t$$

$$t = \frac{1}{\lambda}\, ln\frac{N_0}{N}$$

$$t = \frac{1}{4.95 \times 10^{-11}}\, ln\left(\frac{10}{9}\right)$$

$\therefore$ $$t = 2.10 \times 10^9\ years\ (\boldsymbol{Ans})$$

***Problem-5:*** *The half-life of radon is 3.82 days. After how many days will only 5% of radon left over?*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{3.82}$$

$\therefore$ $$\lambda = 0.1814\ per\ day$$

Here,

$$N = \frac{5}{100} N_0$$

$$\frac{N_0}{N} = \frac{100}{5} = 20$$

$$T_{1/2} = 3.82\ days$$

$$t = ?$$

We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln\frac{N_0}{N} = \lambda t$$

$$t = \frac{1}{\lambda}\, ln\frac{N_0}{N}$$

$$t = \frac{1}{0.1814}\, ln\, 20$$

$\therefore$ $$t = 16.51\ days\ (\boldsymbol{Ans})$$

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003

***Problem-6:*** *The disintegration constant of a radioactive element is 0.0693 per month. Calculate the required for 75% at the element to decay.*

***Solution:*** We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln\frac{N_0}{N} = \lambda t$$

$$t = \frac{1}{\lambda}\, ln\frac{N_0}{N}$$

$$t = \frac{1}{0.0693} ln\, 4$$

$\therefore$ $$t = 20\ months\ (\boldsymbol{Ans})$$

Here,

As 75% element decays, 25% the element left behind

$$N = \frac{25}{100} N_0$$

$$\frac{N_0}{N} = \frac{100}{25} = 4$$

$$\lambda = 0.0693\ per\ month$$

$$t = ?$$

***Problem-7:*** *Find the activity of 1.00 mg of radon, $^{222}$Rn, whose atomic mass is 222 U.*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{3.30 \times 10^5}$$

$\therefore$ $$\lambda = 2.1 \times 10^{-6}\ sec^{-1}$$

Here,

$$T_{1/2} = 3.82\ days$$

$$= (3.82 \times 24 \times 3600)\ sec$$

$$= 3.30 \times 10^5\ sec$$

The number of $N$ atoms in 1.00 mg of $^{222}$Rn is

$$N = \frac{1 \times 10^{-3}\ gm}{222\ (gm/molecule)} \times 6.023 \times 10^{23} (atom/molecule)$$

$$N = 2.71 \times 10^{18}\ atoms$$

$$Activity = \lambda N = 2.1 \times 10^{-6} \times 2.71 \times 10^{18}$$

$$= 5.69 \times 10^{12}\ decays/sec$$

$$= 5.69 \times 10^{12}\ Bq\ (\boldsymbol{Ans})$$

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003

***Problem-8:*** *A radioactive substance has a half-life period of 30 days. Calculate*
*(i) The radioactive disintegration constant.*
*(ii) The average life*
*(iii) The time taken for 3/4 of the original number of atoms to disintegrate.*
*(iv) The time for 1/8 of original number of atoms to remain unchanged.*

***Solution:*** (i) We know,

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{30}$$

$\therefore$ $$\lambda = 0.0231\ per\ day\ (\boldsymbol{Ans})$$

Here,

$T_{1/2} = 30\ days$

For the 1st case

$$N = \left(1 - \frac{3}{4}\right) N_0$$

$$N = \frac{1}{4} N_0$$

$$\frac{N_0}{N} = 4$$

$t_1 = ?$

For the 2nd case

$$N = \frac{1}{8} N_0$$

$$\frac{N_0}{N} = 8$$

$t_2 = ?$

(ii) Average life

$$\tau = \frac{1}{\lambda} = \frac{1}{0.0231} = 43.29\ days\ (\boldsymbol{Ans})$$

(iii) We know,

$$N = N_0 e^{-\lambda t_1}$$

$$\frac{N_0}{N} = e^{\lambda t_1}$$

$$ln\frac{N_0}{N} = \lambda t_1$$

$$t_1 = \frac{1}{\lambda}\ ln\frac{N_0}{N}$$

$$t_1 = \frac{1}{0.0231} ln\, 4$$

$\therefore$ $$t_1 = 60\ days\ (\boldsymbol{Ans})$$

(iv) We know,

$$N = N_0 e^{-\lambda t_2}$$

$$\frac{N_0}{N} = e^{\lambda t_2}$$

$$ln\frac{N_0}{N} = \lambda t_2$$

$$t_2 = \frac{1}{\lambda}\ ln\frac{N_0}{N}$$

$$t_2 = \frac{1}{0.0231} ln\, 8$$

$\therefore$ $$t_2 = 90\ days\ (\boldsymbol{Ans})$$

***Problem-9:*** *1 gm of radium is reduced by 2.1 mg in 5 years by alpha decay. Calculate the half-life period of radium.*

***Solution:*** We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln\frac{N_0}{N} = \lambda t$$

$$\lambda = \frac{1}{t}\ ln\frac{N_0}{N}$$

$$\lambda = \frac{1}{5} ln\left(\frac{1}{0.9979}\right)$$

$\therefore$ $$\lambda = 4.20 \times 10^{-4}\ \ per\ year$$

Here,

$N_0 = 1\ gm$

$N = (1 - 2.1 \times 10^{-3}) gm$

$N = 0.9979\ gm$

$\frac{N_0}{N} = \frac{1}{0.9979}$

$t = 5\ years$

$T_{1/2} = ?$

We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$T_{1/2} = \frac{0.693}{4.20 \times 10^{-4}}$$

$\therefore$ $$T_{1/2} = 1650\ year(\boldsymbol{Ans})$$

***Problem-10:*** *A radioactive sample has its half-life equal to 60 days. Calculate the time required for 2/3 of the original number of atoms to disintegrate.*

***Solution:*** We know, Half-life

$$T_{1/2} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{0.693}{T_{1/2}}$$

$$\lambda = \frac{0.693}{60}$$

$\therefore$ $$\lambda = 0.01155\ per\ day$$

Here,

$N = \left(1 - \frac{2}{3}\right) N_0$

$N = \frac{1}{3} N_0$

$\frac{N_0}{N} = 3$

$T_{1/2} = 60\ days$

$t = ?$

We know,

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\lambda t}$$

$$\frac{N_0}{N} = e^{\lambda t}$$

$$ln\frac{N_0}{N} = \lambda t$$

$$t = \frac{1}{\lambda}\, ln\frac{N_0}{N}$$

$$t = \frac{1}{0.01155}\, ln\, 3$$

$\therefore$ $t = 95.11\ days\ (\boldsymbol{Ans})$

Prepared by- Md. Matiur Rahman, Lecturer (Dept. of Medical Physics), KYAU
Mail: mrroyal2006@live.com, Mobile No.: 01723349003